

Tradition, Science, and the Search for Truth

Foreword by Prof. Sayam Sen Gupta

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First, my warmest congratulations to the team behind *InSight*. Running a scientific magazine is an experiment in itself – one that demands curiosity, persistence, and no small measure of courage. At a time when most information reaches us through fleeting WhatsApp forwards and endless social media reels, it is deeply heartening to see students choosing the slower, more thoughtful medium of print. There is a unique joy in telling the stories behind scientific discoveries, and *InSight* gives those stories a home. For that, our entire community owes its editors and contributors sincere gratitude.

When the editorial team invited me to write this piece, I found myself wondering what message would best suit the first issue. One question that repeatedly surfaces today is whether science and tradition stand in opposition. It is a fascinating debate, and one that deserves a closer look.

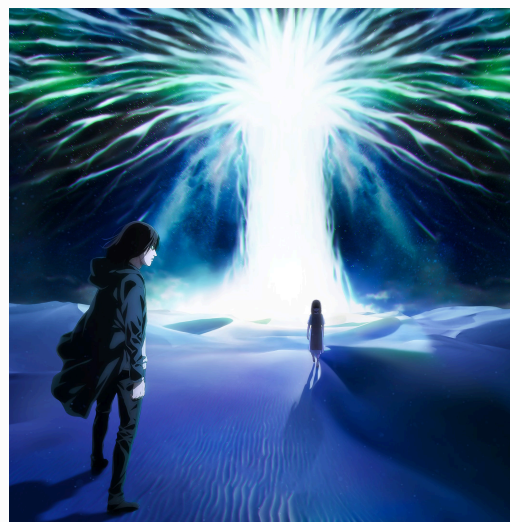
Consider the story of beriberi, a disease that once devastated communities across Asia after steam mills began polishing rice, removing its nutrient-rich outer layers. Long before Christiaan Eijkman demonstrated the role of thiamine through careful experiments in the late nineteenth century, many South Asian communities had already developed the practice of parboiling paddy before milling. This simple process – still familiar to us as “seddho chaal” in Bengal and in the preparation of rice for foods such as idli and dosa – preserved much of the vitamin within the grain. The same pattern appears elsewhere. After screening more than 240,000 compounds without success, Chinese scientist Tu Youyou turned to traditional Chinese medical texts for inspiration that were used for over two millennia to treat fevers. Her team identified sweet wormwood (*Artemisia annua*) as a treatment for intermittent fevers and, in 1971, isolated artemisinin, its potent antimalarial compound. Artemisinin-based therapies are now the WHO’s standard treatment for malaria, and Tu Youyou received the 2015 Nobel Prize in Medicine.

None of these communities for centuries knew about thiamine, or artemisinin. What they possessed was something equally valuable: knowledge accumulated through generations of observation, experimentation, and lived experience. Tradition, in this sense, is not superstition. It is a repository of successful solutions refined over centuries. But tradition has one limitation – it evolves slowly. When the Industrial Revolution transformed food production and populations grew rapidly, longstanding practices could not adapt quickly enough. This was not a failure of tradition; it was a mismatch of timescales.

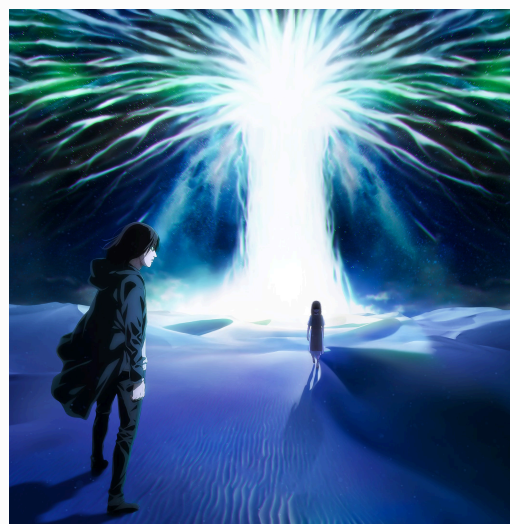
That is where hypothesis-driven science becomes indispensable – not as a replacement for tradition, but as its accelerant. What once required centuries of trial and error can now be understood within a generation through careful experimentation, mechanistic insight, and peer review. The COVID-19 pandemic offered perhaps the clearest demonstration. Within weeks of the first reported cases, scientists had identified the virus and sequenced its genome. Within months, they had determined the structure of its spike protein. Within about a year, that knowledge had culminated in safe and effective vaccines developed at an unprecedented speed.

Knowledge accumulated slowly across generations and knowledge built rapidly through scientific inquiry are not competing ways of understanding the world. They are complementary. One reminds us where wisdom comes from; the other shows us how quickly that wisdom can grow when guided by evidence.

May *InSight* continue to tell both kinds of stories.



This article is *extremely* educational. Be sure to read it ... [Read the rest here.](#)



This interview is *extremely* interesting. Be sure to read it ... [Read the rest here.](#)

A Word from the Editors

Debanuj Chatterjee

Editor at InSight, IISER Kolkata

Every scientific breakthrough begins with a question. Some questions seek to understand the universe at its most fundamental level, while others emerge from observing the natural world around us or reflecting on the journeys that shape scientific careers. Despite their diversity, they are united by a common thread: curiosity. It is this curiosity that continues to drive science forward, and manifests as the anthem of InSight.

We begin at the fascinating intersection of theoretical physics and mathematics: the world of conformal bootstraps. A remarkable blend of physics, geometry, and mathematical consistency forms the foundation of this field. The journey explores how quantum mechanics can be demystified by demanding the self-consistency of nature.

Gears change as we delve into the world of chemistry. Our next article explores the development of asymmetric synthesis and modern hybrid catalytic systems. The ability to selectively produce one molecular form over another has been transforming pharmaceutical science and chemical engineering over the years. This underscores how these discoveries are fundamentally reshaping modern medicine.

This year marks one hundred years since Werner Heisenberg's formulation of matrix mechanics, a milestone that forever changed our understanding of reality. Our feature on "Quantum at 100" revisits the revolutionary ideas that gave birth to quantum mechanics and reflects on how they continue to influence modern science, ranging from condensed matter physics to quantum information theory to materials science.

Science, however, is not confined to laboratories or equations alone. In a deeply personal narrative, we follow the journey of a young scientist whose dream of studying at IISER Berhampur unfolded through uncertainty, perseverance, and probability itself. It is a reminder that every scientific career is built not only on intellect, but also on resilience and the courage to embrace the unknown.

We also travel millions of years into the Earth's past to uncover the mystery behind the disappearance of giant dragonflies. Their story offers a fascinating glimpse into how changes in atmospheric composition and planetary environments have shaped the evolution of life on Earth.

Whether you are a student, researcher, or simply a curious reader, we hope this issue will spark new ideas and conversations. As we stand at a crucial crossroads of Indian science and education, we hope the growing InSight community continues to champion scientific thinking. The future of science depends not only on discoveries in laboratories but also on a society willing to ask questions. Afterall every scientific breakthrough begins with a question.



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An Example Title: Story of my x^2

Author Name **Institute**

Dec 07, 2025

Reviewed By: Reviewer 2

This article is extremely educational. Be sure to read it "carefully".



Author description. Might be important to use formatting here, for example when stating the name of the author.

Initial paragraph. Everything allowed in later paragraphs (to be described) can be used here as well. The first paragraph must be passed into the `dcap` function to insert a drop cap at the beginning of the article.

Section Headings

Now add **content** as *usual*, formatting as necessary.

Add images:

To add quoted statement from someone, or a line of poetry, etc, use the #quote function:

This is a quote from someone famous

Can also be used for something *meta*, such as dedicating the article to someone at the beginning:

Dedicated to no one...



FIG 1 : *Example image. This image only spans one column because portrait is to true. position:top forces it to the top of the page.*

Math is inserted by enclosing within backticks. Inline math: $x \rightarrow y$.

Block math:

$$\pi = \int_0^{\pi} \tan(x) \, dx$$

Multiline block math:

$$\begin{aligned} \langle p \rangle &= \text{expec. value} \\ \Sigma &= \lim_{x \rightarrow 0} \lim_{y \rightarrow \infty} \lim_{z \rightarrow \frac{1}{\infty}} f(x^\dagger * y^z + \frac{z}{x}) \\ &\quad + \lim_{y \rightarrow 0} \lim_{z \rightarrow \infty} \lim_{x \rightarrow \frac{1}{\infty}} f(z^\dagger * x^y + \frac{1}{z}) \end{aligned}$$

Footnotes can be added:¹

References can be added, either as [1], [2], or as ¹, ². References must be added in the file mentioned at the top of the file in order for them to appear in the article.

Lists can be added easily. Unordered list:

- item 1
- item 2

Ordered list:

1. item 1
2. item 2

A #colbreak forces a column break at this point:

¹This appears in the footer.



FIG 2 : Example image. This image body spans both columns because portrait is to false and width is > 66%. position: auto means it will be positioned wherever is comfortable.

The present text will appear on the other column of the page. Use this to distribute content evenly on the two columns.

References

1. Ref 1
2. Ref 2



FIG 3: Example image. This image + caption combined spans both columns because portrait is to false but width is <= 66%. position: bottom forces it to bottom of page.



An Example Interview: Monster meets monster

interview by

Big Brother

Institute of Xerxes

African Swordsman

Apocalyptic Forest

 December 07, 2025

This interview is extremely interesting. Be sure to read it "carefully".



Interviewee 1 description. Might be important to use formatting here, for example when stating the name of the author.

Interviewee 2 description

BB: Welcome. I'm Big Brother, state alchemist and full-time short person with attitude. Today we're interviewing two individuals with... complicated relationships with global peace.

AS: I'm Afrcan Swordsman. I hit things. Let's make this quick.

BB: Our guests: Hobo Eager and Lulu Lampard. Try not to overthrow anything during the interview.

HE: No promises.

LL: I only overthrow when strategically necessary.

BB: First question. On a scale from one to "world-ending catastrophe," how far are you willing to go for your ideals?

HE: Define catastrophe.

AS: That's already a bad sign.

LL: Hypothetically, if reshaping the world eliminates suffering, then temporary catastrophe is simply an investment.

HE: Exactly.

BB: I regret inviting both of you.

AS: You two talk about changing the world. Ever try just swinging a sword at your problems?

HE: I prefer overwhelming force on a continental scale.

LL: I prefer chess.

AS: So one of you tramples everything. The other monologues until it collapses.

LL: That is a gross oversimplification.

AS: Works, though?

LL: ...Occasionally.

AS: Final question. Do you ever worry you've become the villain?

HE: If I am, it's for a reason.

LL: History is written by the victorious.

AS: Hm.

BB: That's not comforting.

AS: No. But it's honest.



FIG 1 : Example image. This image only spans one column because portrait is to true. position:top forces it to the top of the page.

BB: Well, thank you for not destroying the studio.

HE: Yet.

LL: I calculated the odds. It wasn't efficient.

AS: Good. Because I left my sword outside, and I'd rather not fetch it.

BB: And that concludes today's interview before someone declares war on the furniture.